Asphalt

Advanced Performance Tests

The GCTS provides modular testing systems that allows for a wide variety of asphalt performance tests to be performed. This system is capable of performing

- Dynamic Complex Modulus
- Flow number
- Flow time
- Indirect tension
- Beam fatigue
- Resilient modulus & Direct tension tests.

The Asphalt Pavement Tester features advanced software to allows for accurate results and a completely "turn-key" system. Meets all AASHTO, ASTM, and EN standards.

Dynamic Complex Modulus test

The Dynamic Modulus [E*], the absolute value of the complex modulus, is used to predict the rutting potential of HMA.The GCTS system performs this test with ease in both unconfined and confinded condition, and automatically obtain the master curve.

This program includes a function solver to obtain the Master Curve from the tesdats according to AASHTO or Witczak functions.

Indirect Tension and Creep Compliance Test

The tensile strength of HMA is a major predictor of the service life of pavement. The cracking potential of the HMA is directly related to its tensile strength.

Beam Fatigue Test

The GCTS Beam Fatigue Testing System can be used to determine the fatigue life of a layer of Hot Mix Asphalt. This system is controlled by the advanced software developed by GCTS and is a completely "turn-key" system. The software automatically calculates multiple fatigue life parameters, including initial stiffness, maximum tensile stress and strain, flexural stiffness, phase angle, dissipated energy per cycles, and number of cycles to failure.

Resilient Modulus Test

The GCTS Resilient Modulus Testing System features a modular design that allows for various tests to be performed using a single machine. This allows many asphalt and soil parameters to be calculated with this device. This system is completely "turn-key," so once the specimen has been prepared, all testing procedures are computer controlled.

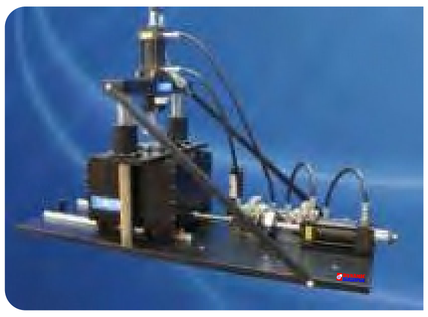
Environmental Chamber

The GCTS Environmental chamber allows for testing at a specific temperature by either cooling or heating in a wide temperature and with a better than +/-0.5 C precision, using Integrated PID digital control through GCTS application software. Includes high air turnover for maintaining uniform temperatures throughout the chamber, Full size heated multi-panel front window to prevent fogging, and LN2 port and spray nozzle for cooling.

Asphalt & Pavement

Direct Shear Test

The GCTS Direct Shear Testing Systems can perform both constant normal stress and constant normal stiffness direct shear tests on either asphalt or soil specimens. These systems feature closed-loop servo control of the shear load and normal load actuators, and are completely "turn-key" systems. Through advanced software and hardware, these systems essentially eliminates all vibrations that would disturb specimen during testing.



Hamburg Wheel Tracker

The GCTS Hamburg Wheel-Tracking Tester is used to test submerged, compacted Hot Mix or Warm Mix Asphalt specimens as per AASHTO T 324 standard. The device consists of a reciprocating wheel that is rolled over the specimens while measuring the rate of permanent deformation. A constant load of 705 N is applied to the specimens with a sinusoidal wheel speed of 1 ft/sec in a constant temperature controlled water bath. An LVDT jig is used for continuous measurement of permanent deformation along the path of the rolling wheel.

Roller Compactor cum Wheel tracker

Electro-pneumatically operated machine which combines two Functions, one for preparing

Slabs made of bituminous mix (roller compaction) and one for wheel tracking.

The roller compaction function Is based on the "kneading" Method using sliding metal plates aligned vertically in the mould while the wheel Tracking function is in compliance with the "small-size" method of EN 12697-22 and EN12697-33

Vertical movement is obtained pneumatically and horizontal movement is obtained using a Brushless motor.

SHRP Test Equipment Range

Pressure Ageing Vessel & Vacuum Degassing Oven

The PAV3 pressure aging vessel is used to simulate in-service oxidative aging of asphalt binder according to procedures developed by the Strategic Highway Research Program (SHRP). The PAV consists of a vertical, stainless steel pressure vessel in a stainless steel cabinet with encased band heaters, a precision sample holder for simultaneous testing of ten specimens, a set of ten TFOT specimen trays, a pressure controller, temperature controller, pressure and temperature measurement devices, temperature and pressure recorder, and a specimen loading and unloading tool. The vacuum degassing oven (VDO) is used to precisely and accurately degas pressure-aged bindersamples to meet the standards. The compact, table-top unit is constructed of stainless steelwith a hinged lid to conserve space while allowing easy access to the stainless steel vacuum chamber.

Rolling Thin Film Oven Test

The rolling thin film oven is used to measure the effect of heat and air on a moving film of semisolid asphaltic material and is an indicator of the approximate change in properties during conventional hot-mixing. The results of this treatment are determined from measurements of the asphalt properties before and after the test.

Asphalt & Pavement

Loss-on-Heat Thin Film Oven

This dual-purpose 16 cuft oven can be used for loss of heat test and thin film test for bitumen and asphaltic materials. The loss-on-heat oven determines the effect on asphaltic materials of heating in an oven under prescribed conditions. The results are reported in terms of change in sample mass and/or changes in selected properties such as viscosity, penetration and ductility as evidenced by test data taken before and after the oven cycle.

Rotational Viscometer and Rheometer

The rotational viscometer and Thermosphere with a precision temperature controller provide an excellent solution for those looking for SHRP asphalt binder testing equipment specifically designed to meet the requirements of AASHTO T316/ASTM D4402 high-temperature test methods. The instrument provides a measuring range of 20-6,000,000 cP with 54 speeds in a range from 0.01 to 200 rpm. A graphic display delivers clear data readings and an intuitive full-touch keypad provides efficient data selection.

Bending Beam Rheometer

The bending beam rheometer (BBR) performs flexural tests on asphalt binder and similar specimens per tests, initially developed by the Strategic Highway Research Program (SHRP), which consist of a constant force being applied to a specimen in a chilled fluid bath in order to derive specific rates of deformation at various temperatures. The complete BBR system consists of a fluid bath base unit, a three-point bend test apparatus, which is easily removed from the base unit for specimen loadingand unloading, an external cooling unit with temperature controller and a calibration hardware kit with carrying case.

Gyratory Compactor

Superpave Gyratory Compactor is a completely self-contained SGC that is ideal for on-site testing and mobile labs, this SGC can handle design and QC/QA work with equal ease. The superpave gyratory compactor measures the mold angle during compaction. This angle, along with the consolidation pressure, gyration number, and specimen height is displayed throughout the compaction process. Simply enter the compaction parameters, lower the prepared mold into the compaction chamber, securethe gyratory head, and press Start. The system then applies the consolidation pressure, induces the angle, and gyrates the mold until the specified number of gyrations or specified height is reached. Extrude the specimen after compaction with the same hydraulic ram used to compact the specimen.

Gyratory Compactor - Servo Controlled

The GCTS Superpave Gyratory Compactor is expertly designed to yield highly accurate results in the variety of tests it can perform.

The introduction of a self-heating mold with precise temperature control makes this system extremely easy to use with only the push of a button.

Continuous positive feedback to maintain gyration angle precision throughout compaction with Closed-Loop servo control of ram pressure and tilt angle.Automatic material mass measurement. Measures shear stress.

Asphalt & Pavement

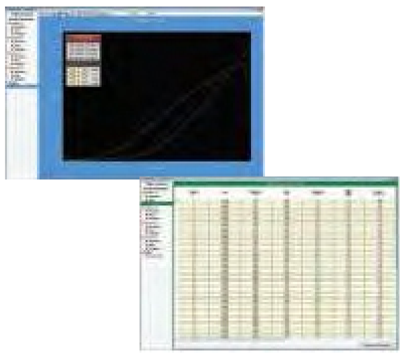
Marshal Stability Test

Automated/ Computerized Loading Frames:

The digital MasterLoader provides the ultimate solution for a lab looking to perform Marshall and Hveem testing, but would also like to be able to use the load frame for all their other testing needs as well as soil tests such as CBR and triaxial tests including UU, CU, CD and UC. The digital MasterLoader has the ability to work as a stand-alone unit, which can perform Marshall tests at the push of a button; or with the aid of Humboldt's Marshall master software and a computer, it can be automated to run tests and gather data in real-time data acquisition in the form of charts and graphs.

Humboldt Marshall Master Software:

Humboldt Marshall master reporting softwareis a stand-alone application used in conjunction with Humboldt's load frames, miniLoggers, etc. to provide real-time data acquisition. The Marshall software provides a simple, test-specific interface to control Marshall test operations and automatically record data while also displaying it in real-time tables and graphs.



Automatic Compactors:

Humboldt automatic, Marshall compaction machines are designed to provide a stable and rigid mechanism for producing 4" or 6" diameter asphalt pills used in Marshall tests. These Marshall compaction machines are available in two types of configurations: one with a rotating mold with a tapered-foot hammer assembly, and the other, a stationary mold with a flat-foot hammer.

Water Baths:

Humboldt water baths feature a microprocessor-based digital controller for precise temperature control throughout their temperature range.

Asphalt/ Binder Content Measurement

Ignition Method

The asphalt content/binder furnace with internal automatic balance is an environmentally-friendly and cost-effective method for the accurate determination of asphalt content. Developed by NCAT, the National Center for Asphalt Technology, USA. The furnace's large capacity handles samples up to 5,000 grams. Ignition method reduces testing time compared to solvent testing methods and automatic operation frees technicians for other tasks.

Asphalt & Pavement

Centrifuge Extractor,
Filterless

Centrifuge Extractor (Explosion Proof):

The asphalt centrifuge extractor is designed for determining the percentage of bitumen in asphaltic mixtures. These extractors conform to the explosion-proof standards for the safety of operating personnel.

Centrifuge Extractor, Filterless

The continuous-flow filterless centrifuge extractor is ideally suited for use in the extraction of mineral fines from bitumen-laden solvents obtained from standard asphalt extraction tests. The system allows the continuous feeding of the suspension until the solids retaining capacity of the beaker is reached.

Vacuum Extractor

Basic vacuum extractor for use in quantitative determinations of bitumen in hot-mixed paving mixtures and pavement samples. Provides a 305mm dia filtering surface. Use with a 4,000cc erlenmeyer flask and a vacuum pump.

Ductility Test

Ductility Machines:

Humboldt ductility machines determine ductility of formed asphalt/cement or semi-solid bitumen by measuring the distance of elongation before reaching the breaking point of a briquette sample, which is pulled apart at a specific speed and temperature. Available as three-speed machines designed for standard and force ductility tests. Tests three briquettes simultaneously and its DC, direct-drive motor maintains constant speed, entirely vibration-free

Temperature Controller:

The Circulating Temperature Controller is designed for use with the Humboldt ductility machine. It provides a solid-state, thermostatically controlled bath and circulator to maintain water temperature within a 0.5°C.

Force Determination Adapter:

Provides precise tensile strength measurement of any material, preparation, procedure or type of test to an accuracy of 0.01 lbs. Attaches over existing pin in ductilometer without tools or machine modi?cation, eliminating need for dedicated equipment for standard and force ductility testing.

Vacuum Pycnometer (Rice) Test

Used to determine the maximum specific gravity of bituminous paving mixtures with maximum aggregate size up to 19.1mm.

Penetrometers

Universal Penetrometer

Direct-reading instrument for precision penetration measurements of bituminous materials, cement, petroleum and waxes, as well as food, cosmetics and pharmaceutical products. Unit has 5" diameter indicator dial, graduated in 400 divisions of 0.1mm, corresponding to 40mm penetration.

Penetrometer for Battery Paste

Combines the universal penetrometer and a grease cone, which has a hardened steel tip with a special plunger.

Penetration Needles and Cones

A wide range of penetration needles and cones are available that can be used with the Universal Penetrometer to conduct penetration tests on different materials

Penetration Needles and Cones

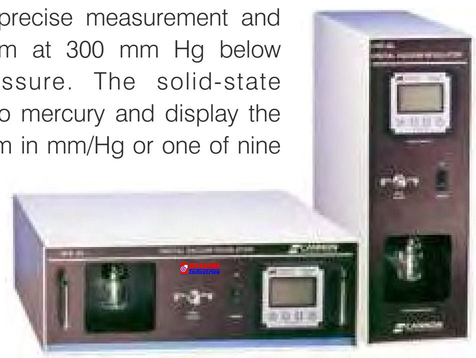
Viscosity Measurement

Kinematic Viscosity Measurement:

Constant temperature baths to maintain accurate control required by ASTM D445 for kinematic viscosity measurements with glass capillary viscometers.

Digital Vacuum Pressure Regulators

These digital vacuum regulators (DVR) aredesigned for precise measurement and control of vacuum at 300 mm Hg below atmosphericpressure. The solid-state Regulators use no mercury and display the amount of vacuum in mm/Hg or one of nine other units of measurement selected using a keypad on the meter.



Saybolt Viscosity Bath

Designed for Saybolt universal and furol viscosity testing, this constant temperature bath meets all ASTM and AASHTO requirements for precise temperature control. The micro-processor PID circuitry assures accurate temperature control within ASTM tolerances throughout the range of ambient to 240°C. At the push of a button, the automatic timer starts the sample flow, senses the 60mL end point and digitally records and displays the efflux time in 0.1 seconds resolution with an accuracy of 0.05%.

Pavement

Compaction Measurement

Nuclear Density Gauge (NDG)

Moisture/Density Gaugesuse an advanced micro-processor- based technology to provide highly-accurate measurements of density and moisture that are automatically computed for direct readouts of wet density, dry density, moisture content, percent of moisture, percent of compaction (Proctor or Marshall), void ratio and air voids. The NDG can be used to test the degree of compaction of practically any paving material such as soil, aggregate, asphalt, concrete, even flyash.

Manufacturer approved re-calibration facilities are available to ensure that gauge accuracy is maintained over several years of use.

Electrical Density Gauge (EDG)

The Electrical Density Gauge (EDG) is a nuclear-free alternative for determining the moisture anddensity of compacted soils used in road beds andfoundations. The EDG is a portable, battery-poweredinstrument capable of being used anywherewithout the concerns and regulations associatedwith nuclear safety. Its user-friendly, step-by-stepmenu guides the user through each step of thetesting procedure and cautions the user when valuesdo not correspond to established curves forthe material being tested.

Easy-to-use, the EDG can be used as a constructionaid to monitor day-to-day compaction operationsby providing performance and measurementresults highly comparable to those achieved withtraditional methods, including the nuclear gaugeand/or a sand-cone and oven moisture test combination.

Strength Measurement

Electronic Plate Bearing Tester - Static

The Electronic Plate Load Device allows simple determination of the load-bearing and deformation capacity of soils. The load settlement lines and the modulus of deformation Ev1, Ev2 are determined to DIN 18134:2001-09 or E DIN 18134:2010-04 (plate load test). Thanks to the splashproof housing, external buttons and a backlit display, it can also be used under inclement ambient conditions. Test logs can be printed out directly on site. The results saved on an SD card during the test can be transferred to the PC and are available for further processing under Microsoft Excel®

Asphalt & Pavement

Dynamic Light Weight Deflectometer/ Drop Weight Tester

This dynamic plate load test is used in earthworks, civil engineering, road construction and rail track constructionto determine the load-bearing capacity and to assess the compaction of soils, loose base courses and soil improvement in accordance with TP BF – StB Part B 8.3.

The test method is suitable for course-grained and mixed-grained soils up to a maximum grain size of 63 mm. The test result is displayed in MN/m². The measuring range of the light drop weight tester lies between 15 and 70 MN/m². With the 1.5x impact load achieved with the 15 kg loading device, the measuring range is extended to 70 – 105 MN/m². This is required for heavy base courses in road beds and for ballast.

Dynamic Light Weight Deflectometer/ Drop Weight Tester - For Asphalt Pavements

Similar in operation as the Dynamic Light Weight Deflectometer discussed above but with stamps of small diameters for measurement on Asphalt pavement layers. Extremely useful for quick quality control during the asphalt layer compaction process.

Geogauge

The GeoGauge is a unique, QC/QA field tool that can be used to measure the uniformity of unbound pavement layers by measuring the variability in stiffness throughout a structure. It is an excellent tool for identifying construction anomalies that would otherwise go undetected during construction where only density or percent compaction measurements were used. GeoGauge can be used to verify that the stiffness/ modulus values assumed in the design specifications of new or rehabilitated pavement structures are met.

This dynamic technology used by the GeoGauge simulates real in-use conditions. This factor allows the GeoGauge to directly measure in-place engineering properties during the construction process.

Benkelman Beam

The Benkelman beam measures the deflection of a flexible pavement under moving wheel loads. Extremely accurate and easy to use. Direct-reading dial indicator eliminates need for conversion table or field calculations. This pavement analysis tool provides precision accuracy and it is a light weight tool where no needs for conversions or field calculations, gauges are direct read

Skid Resistance, Polishing

Portable Skid Resistance Tester

Measures friction (skid resistance) on flat, cambered,or gradient road surfaces. Originally developedby the Transport and Road Research Laboratoryof Great Britain.A pendulum arm, having a spring-loaded rubber slider on the pendulum foot. Device is placed on the portion of the road surface to be tested. It is then leveled, and the height of the center of suspension of the pendulum is adjusted to a fixed value which is read on a special gauge. The pendulum is then released from its horizontal position, to swing down freely until the rubber slider contacts the test surface. As the slider travels across the surface for a fixed distance, the pendulum is slowed and a frictionally-constrained pointer affixed to the pendulum arm measures the highest point in the pendulum arc. The position of the pointer is then read on a measuring arc graduated from 0 to 150. Pointer readings indicate the resistance to skidding of the test surface. Compact, easily transported and used in thefield. Can be used in remote locations, independent of any vehicle. Enclosed bearings & working parts for protection against wear and contamination

Accelerated Polishing Machine

The accelerated polishing machine polishes samplesof aggregates, simulating actual road conditions,and is used in conjunction with the portableskid resistance tester to determine the polishedstone value for aggregates used in road surfaces,and gives a measure of the resistance to skidding.Fourteen specimens are clamped aroundthe "road wheel" of the polishing machine, andsubjected to two stages of polishing by a loadedrubber tire. First by corn emery, and secondlyby emery flour. Use of the accelerated polishingmachine and the skid resistance tester inroad construction has had a major influence inreduction of accidents.

Asphalt & Pavement

Pavement Strength/ Design Evaluation

Falling Weight Deflectometer

The Falling Weight Deflectometer is used to measure the vertical deflection response of a surface to an impulse load. Precision load measurement and deflection sensors measure and record the pavement surface characteristics that are used to calculate pavement properties, such as Bearing capacity, Layer thickness, E moduli, Expected surface life.

This non-destructive testing device can determine the actual material used and in what combination to build the pavement surface. It also identifies voids underneath the surface or how two surfaces (typically concrete slabs) are in contact with each other.

Available as Standard Falling Weight Deflectometer and Heavy Falling Weight Deflectometer.

Traffic Speed Deflectometer

The Traffic Speed Deflectometer, TSD is a rolling wheel deflectometer for network level bearing capacity measurement.

Using a patented Doppler technology it is possible to measure the pavement response to an applied 10 – ton axle load travelling at traffic speed of up to 80 km/h. TSD data provides continuous pavement deflection profiles, from which bearing capacity indices can be derived and pavement fatique / residual life can be estimated. TSD is the ideal tool to measure the bearing capacity of large road networks. The high accuracy and resolution makes TSD perfect to pin point locations with bearing capacity deviations.

Detailed knowledge of the network bearing capacity minimize the need for traditional stationary or slow moving, discrete measuring techniques.

Roughness Index/ Laser Profiling

LaserProf

LaserProf is an inertial profiler designed for quality control and supervision of highways. It enables the user to measure longitudinal profiles of any pavement surface. Derived results from the pavement profiles such as IRI can be calculated.

LaserProf can be configured with Texture Sensors & GPS. The system is portable and easily mountable on any vehicle.LaserProf enables the measurement at normal traffic speed, with maximum speeds exceeding 150km/h.

Profilograph

Profilograph is a specially designed system for transversal and longitudinal profile combined in one 3D profile.

The system is equipped with high precision sensors, digital data acquisition and processing for highest possible result quality.

The system is configurable to meet the customer's requirements. It can either be configured as a standalone system or part of a larger multi-function vehicle. The Profiles can be synchronized with other measurement systems like GPS, Surface Imaging & ROW (Right of Way) imaging.

High-Low Detector (Rolling Straight Edge)

Basic instrument to measure planeness of pavement surfaces, such as highways, airport runways, bridge decks, etc. Requires only one operator to detect, register, and dye mark high and low areas that need to be ground down or filled. Scraper blades keep wheels clean to maximize accuracy. Includes wheel stands for stabilizing during calibration, transport and storage. There is provision for checking wheel alignment. A steering handle is provided for easy positioning and movement, and provides a button for dye discharge.
Marking dye is used to mark high and low areas on pavement.

Asphalt & Pavement

Surface/ ROW/ Subsurface Imaging Systems

Surface Imaging

Surface Imaging system features line scan camera technology for continuous undistorted images for conditional monitoring of highways.

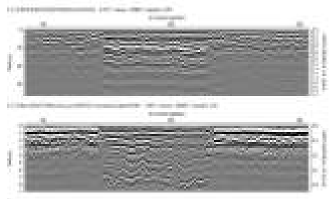
An odometer mounted on the wheel of the vehicle, through control electronics, triggers the camera at fixed intervals. Each line from the camera is added to the previous lines to make up one long continuous image. The continuous image is stored on a computer, tagged with time and distance identifying the precise position for each part of the image. Post processing software synchronize the image with other measurements made in the same vehicle based on the high precision odometer

Right-Of-Way (ROW) Imaging

The Right-of-Way Imaging (ROW) system records and stores images of the surrounding while driving at traffic speeds. The system can operate as a separate system or be a subsystem to a profiling system where it can be synchronized with the profile and with GPS data.

The ROW system can function with one or multiple cameras. There is a wide range of digital cameras for optimum image quality and large flexibility and for the user there is selectable storage format for maximum flexibility.

Subsurface Radar Imaging



GroundVue3 is a multi-channel Ground Penetrating Radar that can be used to determine Pavement Layer Thickness, Pavement subsurface defects, Concrete/ RCC Integrity Testing, Foundation Assessment, Underground Utility detection, Archaeological applications among others. The system is capable for simultaneous multi-channel High Speed Orientation and is capable for towing behind or attachment to a survey vehicle. The system also features Auto-depth calibration option.

Retro-reflectivity Measurement

For Road Marking (Dynamic Retroreflectometers)

The Dynamic Retroreflectometer measures the night visibility (RL) of road markings as seen by a vehicle driver driving with dipped headlight. This allows an objective measurement of retroreflection.

Features :

- Easy and fast attaching and removing of the measuring head of the vehicle.
- The measuring head can be installated on either side of the car
- Markings of different colours can be measured
- Suitable for profile markings
- Also suitable for stationary readings of road markings under continuous wetting using a rain simulator

Portable Handheld Retroreflectometer

The classic portable handheld retroreflectometer with memory for determination of night visibility (RL) and / or day visibility (Qd) of road markings as well as ambient temperature (°C/°F) and relative humidity (rH %)

High resolution color touchscreen with excellent visibility under all lightconditions Ultrafast retroreflection measurement (RL and Qd) For all types of road markings In accordance with EN 1436 (RL/Qd), ASTM E 1710 (RL), ASTM E 2302 (Qd) and ASTM E 2177 (RL wet)

For Traffic Signs and Reflective Clothing

For Determination of night visibility (coefficient of retroreflec¬tion RA and R') of traffic signs, safety garments and other reflexive materials with measurement of three different observation angles at the same time.

The very first retroreflectometer with LED illumination system and with a 3.5" high resolution colour touchscreen with adjustable display inclination for excellent visibility under all light conditions also in bright sunlight.